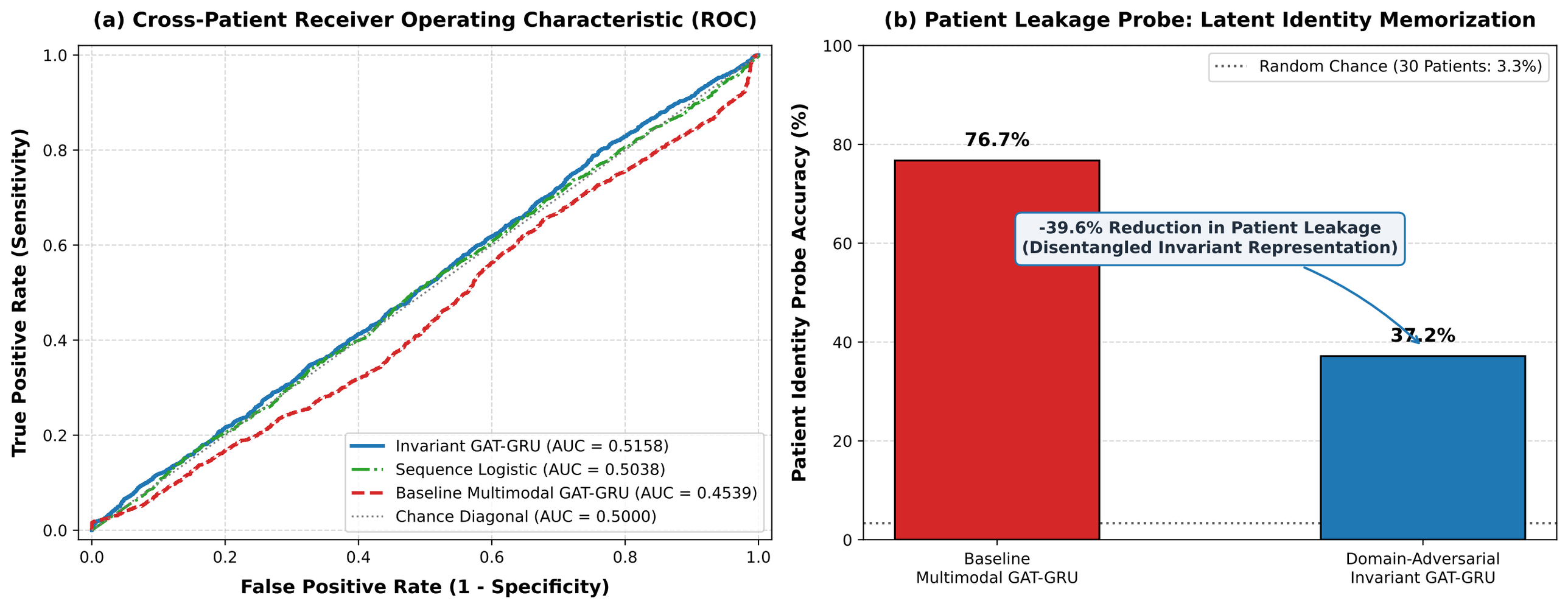

(a) Cross-Patient Receiver Operating Characteristic (ROC)
True Positive Rate (Sensitivity)
False Positive Rate (1 - Specificity)
0.0
0.2
0.4
0.6
0.8
1.0
Invariant GAT-GRU (AUC = 0.5158)
Sequence Logistic (AUC = 0.5038)
Baseline Multimodal GAT-GRU (AUC = 0.4539)
Chance Diagonal (AUC = 0.5000)
(b) Patient Leakage Probe: Latent Identity Memorization
Patient Identity Probe Accuracy (%)
0
20
40
60
80
100
Random Chance (30 Patients: 3.3%)
76.7%
-39.6% Reduction in Patient Leakage
(Disentangled Invariant Representation)
37.2%
Baseline
Multimodal GAT-GRU
Domain-Adversarial
Invariant GAT-GRU